THE DELTA-INVARIANT AT A POINT CAN ONLY GROW UNDER EXTENSION OF THE BASE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites five cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Geometric reducedness can be checked on local rings. Stacks Project, Varieties, Lemma lemma-geometrically-reduced-at-point (tag 035W). Gives geometric reducedness of A, which is what makes B a localization of the base-changed ring.
- The pre-delta construction and the equivalent conditions on the element f. Stacks Project, Varieties, Lemma lemma-pre-delta-invariant (tag 09NL) and Lemma lemma-pre-pre-delta-invariant (tag 09NK). Supplies the intermediate ring A' and the element f whose behaviour is preserved under the field extension.

Source reference. Original lemma 035W: sources/stacks-chow/tags.txt, source lines 4143–4143. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3S: sources/stacks-chow/tags.txt, source lines 15202–15202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C3R: sources/stacks-chow/tags.txt, source lines 15201–15201. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme. Let K/k be a field extension and set $Y = X_K$. Let $y \in Y$ with image $x \in X$. Assume X is geometrically reduced at x and $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) = 1$. Then*

$$\delta\text{-invariant of } X \text{ at } x \leq \delta\text{-invariant of } Y \text{ at } y$$

Proof. Set $A = \mathcal{O}_{X,x}$ and $B = \mathcal{O}_{Y,y}$. By Lemma 035W we see that A is geometrically reduced. Hence B is a localization of $A \otimes_k K$. Let $A \to A'$ be as in Lemma 0C3S. Then

$$B' = B \otimes_{(A \otimes_k K)} (A' \otimes_k K)$$

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

is finite over B and $B \to B'$ induces an isomorphism on total rings of fractions. Namely, pick $f \in \mathfrak{m}_A$ satisfying (1) – (6) of Lemma 0C3R; since $\dim(B) = 1$ we see that $f \in \mathfrak{m}_B$ playes the same role for B and we see that $B_f = B'_f$ because $A_f = A'_f$. Let B'' be the integral closure of B in its total ring of fractions as in Lemma 0C3S. Then $B' \subset B''$. Thus the δ-invariant of Y at y is $\text{length}_B(B''/B)$ and

$$\begin{aligned} \text{length}_B(B''/B) &\geq \text{length}_B(B'/B) \\ &= \text{length}_B((A'/A) \otimes_A B) \\ &= \text{length}_B(B/\mathfrak{m}_A B)\text{length}_A(A'/A) \end{aligned}$$

by Algebra, Lemma 02M1 since $A \to B$ is flat (as a localization of $A \to A \otimes_k K$). Since $\text{length}_A(A'/A)$ is the δ-invariant of X at x and since $\text{length}_B(B/\mathfrak{m}_A B) \geq 1$ the lemma is proved. $\square$